

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Hard

Question

x	y
3	7
k	11
12	n

The table above shows the coordinates of three points on a line in the xy-plane, where k and n are constants. If the slope of the line is 2, what is the value of $k + n$?

Rationale

The correct answer is 30. The slope of a line can be found by using the slope formula, $\frac{y_2 - y_1}{x_2 - x_1}$. It's given that the slope of the line is 2; therefore, $\frac{y_2 - y_1}{x_2 - x_1} = 2$. According to the table, the points $(3, 7)$ and $(k, 11)$ lie on the line. Substituting the coordinates of these points into the equation gives $\frac{11 - 7}{k - 3} = 2$. Multiplying both sides of this equation by $k - 3$ gives $11 - 7 = 2(k - 3)$, or $11 - 7 = 2k - 6$. Solving for k gives $k = 5$. According to the table, the points $(3, 7)$ and $(12, n)$ also lie on the line. Substituting the coordinates of these points into $\frac{y_2 - y_1}{x_2 - x_1} = 2$ gives $\frac{n - 7}{12 - 3} = 2$. Solving for n gives $n = 25$. Therefore, $k + n = 5 + 25$, or 30.